F18CE2 - MV Calculus and Real Analysis B: Solutions 4

1 The Bolzano-Weierstrass theorem is: Every bounded sequence has a convergent subsequence.

(a) If (a_n) is the sequence $6, 8, 6, 8, 6, \cdots$ then (a_n) is not convergent. It has the convergent subsequence $6, 6, 6, \cdots$ (or $8, 8, 8, \cdots$, or $6, 8, 6, 8, 8, 8 \cdots$ or lots of others).

(b) We show that (a_n) is bounded.

$$|a_n| = \frac{3n\left|\cos(2n^2+1)\right|}{n+2} \leqslant \frac{3n}{n+2} \leqslant \frac{3n}{n} = 3,$$

so (a_n) is bounded.

By the Bolzano-Weierstrass theorem, (a_n) has a convergent subsequence.

2 By definition,

$$a_{2n}(x) = 2n\left(x^{\frac{1}{2n}} - 1\right) = 2n\left((x^{\frac{1}{2}})^{\frac{1}{n}} - 1\right) = 2a_n(\sqrt{x}). \tag{1}$$

Since

$$\lim_{n\to\infty} a_{2n}(x) = h(x) \qquad \text{and} \qquad \lim_{n\to\infty} a_n(\sqrt{x}) = h(\sqrt{x}),$$

it follows from taking the limit in (1) that

$$h(x) = 2h(\sqrt{x}).$$

3 $\lim_{x\to\infty} f(x) = L$ if, for each $\epsilon > 0$, there exists X such that

$$|f(x) - L| < \epsilon \quad \text{for all } x \geqslant X$$

Let $\epsilon > 0$. If $x > 0$ then

$$\left|\frac{3x}{x^2+2} - 0\right| = \frac{3x}{x^2+2} < \frac{3x}{x^2} = \frac{3}{x} < \epsilon \quad (\text{if } x > 3/\epsilon).$$

Fix $X > 3/\epsilon$. Then for all $x \geqslant X$,

$$\left|\frac{3x}{x^2+2}\right| < \epsilon.$$

4 $\lim_{x\to\infty} f(x) = \infty$ if, for each $M > 0$, there exists X such that

$$f(x) > M \quad \text{for all } x \geqslant X.$$

Let $M > 0$. Now,

$$x^2 - 3x = x(x-3) \geqslant x, \quad \text{if } x \geqslant 4.$$

Let $X \geqslant \max\{4, M\}$. Then for all $x \geqslant X$,

$$x^2 - 3x \geqslant x > M.$$

5 Definition

$\lim_{x\to a} f(x) = L$ if for every $\epsilon > 0$ there exists $\delta > 0$ such that

$$|f(x) - L| < \epsilon \quad \text{for all } x \text{ with } 0 < |x - a| < \delta.$$

Note. In each of the following proofs we find that when we look at $|f(x) - L|$ we can factorise it into something like

$$|f(x) - L| = |x - a| \times \{\text{something more complicated}\}.$$

We then suppose that $|x-a| < 1$ (there is nothing special about 1, since we can make $|x-a|$ as small as we like), and use this to find a bound, say K, on $\{$something more complicated$\}$, and then, we further make $|x - a|$ small enough to kill off the K term and make the overall RHS $< \epsilon$.

(a) Let $\epsilon > 0$. Now,

$$|4x - 1 - 7| = 4|x - 2| < \epsilon, \quad \text{if } |x - 2| < \frac{\epsilon}{4},$$

so setting $\delta = \frac{\epsilon}{4}\cdot$ we see that

$$|4x - 1 - 7| < \epsilon, \quad \text{if } 0 < |x - 2| < \frac{\epsilon}{4},$$

(b) Let $\epsilon > 0$. Now

$$|x^2 - 36| = |x - 6||x + 6|\,.$$

If $|x - 6| < 1$ then $5 < x < 7$ and $|x + 6| = x + 6 < 13$, so

$$|x^2 - 36| = |x - 6||x + 6| < 13|x - 6|.$$

Next, if $|x - 6| < \frac{\epsilon}{13}$ then

$$|x^2 - 36| =< 13|x - 6| < \epsilon.$$

So, putting $\delta = \min\left\{1, \frac{\epsilon}{13}\right\}$, we see that

$$|x^2 - 36| =< \epsilon, \quad \text{if } 0 < |x - 6| < \delta.$$

(c) Let $\epsilon > 0$. Now

$$x^3 - 3x - 2 = (x - 2)(x^2 + 2x + 1).$$

If $|x - 2| < 1$ then $1 < x < 3$ and $|x^2 + 2x + 1| < 9 + 6 + 1 = 16$, so

$$|x^3 - 3x - 2| < 16|x - 2|.$$

Next, if $|x - 2| < \frac{\epsilon}{16}$ then

$$|x^3 - 3x - 2| < 16|x - 2| < \epsilon.$$

So, putting $\delta = \min\left\{1, \frac{\epsilon}{16}\right\}$, we see that

$$|x^3 - 3x - 2| < \epsilon, \quad \text{if } 0 < |x - 2| < \delta.$$

(d) Let $\epsilon > 0$. Now

$$\left|\frac{x+2}{x+5} - \frac{1}{2}\right| = \left|\frac{x-1}{2(x+5)}\right| = |x-1|\,\frac{1}{2|x+5|}.$$

If $|x-1| < 1$ then $0 < x < 2$ and $10 < 2|x+5| < 14$, so

$$\frac{1}{2|x+5|} < \frac{1}{10}$$

(**note** we have used the lower inequality $10 < 2|x+5|$ here, not the upper one $2|x+5| < 14$ as in the previous examples, since we are turning the inequality upside down).

Putting these together we have

$$\left|\frac{x+2}{x+5} - \frac{1}{2}\right| < |x-1|\,\frac{1}{10}.$$

Next, if $|x-1| < 10\epsilon$ then

$$\left|\frac{x+2}{x+5} - \frac{1}{2}\right| < \epsilon.$$

So, putting $\delta = \min\{1, 10\epsilon\}$ proves the result.

6 Let $\epsilon > 0$. Now,

$$\begin{aligned} |2f(x) + 3g(x) - (2L+3M)| &= |2(f(x)-L) + 3(g(x)-M)| \\ &\leqslant 2|f(x)-L| + 3|g(x)-M|. \end{aligned}$$

By the definition of the limit, there exist $\delta_1 > 0$ and $\delta_2 > 0$ such that

$$|f(x) - L| < \frac{\epsilon}{5} \quad \text{for all } x \text{ with } 0 < |x-a| < \delta_1,$$

$$|g(x) - M| < \frac{\epsilon}{5} \quad \text{for all } x \text{ with } 0 < |x-a| < \delta_2.$$

Let $\delta = \min(\delta_1, \delta_2)$. Then if $0 < |x-a| < \delta$,

$$|2f(x) + 3g(x) - (2L+3M)| \leqslant 2|f(x)-L| + 3|g(x)-M| < \frac{2\epsilon}{5} + \frac{3\epsilon}{5} = \epsilon.$$

7 Let $\epsilon = L/2 > 0$ in the definition of limit. Then there exists $\delta > 0$ such that

$$|f(x) - L| < \frac{L}{2} \quad \text{if } 0 < |x-a| < \delta.$$

Then if $0 < |x-a| < \delta$,

$$f(x) > L - \frac{L}{2} = \frac{L}{2}.$$